

Aryan Sharma

SENIOR AI/ML ENGINEER · IIT Kanpur

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SUMMARY

Senior ML Engineer with nearly **5 years** at Jaguar Land Rover, owning systems across the full **ML lifecycle** (curation, training, deployment, MLOps). Work spans **multi-modal semantic search** and retrieval at scale, distributed training infrastructure adopted team-wide, and camera/LiDAR perception. Strong in vector search, VLMs, RAG & AWS MLOps, with mentoring and cross-team technical leadership

TECHNICAL SKILLS

Search & Retrieval	Embeddings, Vector Search (Qdrant, Faiss), ANN Indexing, Contrastive Learning (CLIP), Multi-modal Retrieval, Relevance and Quality Scoring
ML / Deep Learning	LLMs, VLMs, RAG, Transformers, Computer Vision, 3D Object Detection, Panoptic Segmentation, Distributed & Mixed-Precision Training, PyTorch, TensorFlow, LangChain
MLOps & Infra	Kubernetes (EKS), Ray, Argo Workflows/Events, Docker, Helm, AWS CDK (IaC), CI/CD, OpenTelemetry, Grafana, ClearML, Snyk
Cloud Platforms	AWS (EC2, Fargate, Lambda, Step Functions, SageMaker, S3, EFS, FSx, CloudWatch, Route53)
Data Systems	Qdrant, Faiss, MongoDB, Voxel51 / FiftyOne
Languages	Python, C++, Go, SQL, Bash/Shell, MATLAB
Collaboration	Mentoring, Code Review, Knowledge Sharing, Cross-team Technical Leadership

EXPERIENCE

JAGUAR LAND ROVER

Bengaluru, India

Senior AI/ML Engineer (*Graduate SWE → AI/ML Engineer → Senior AI/ML Engineer*)

Jul 2021 – Present

Scene Search: AI Data Curation & Intelligent Search System

- Contributed to building **Scene Search**, a data-curation and search platform for ADAS perception use cases, cutting target-scenario discovery from **~50 engineer hours to 1.25 hours (97.5% reduction)**
- Built a large-scale pre-processing pipeline using **Argo Workflows & Events** to automate dataset export (data lake to S3 to EFS), **embedding generation**, and quality scoring (blur, exposure, uniqueness, duplicates)
- Developed an advanced **data curation tool** on top of open source Voxel51/FiftyOne, integrating custom plugins
- Implemented embedding generation using **CLIP** and a pre-trained VLM, indexing vectors with **Qdrant and Faiss**, and metadata in MongoDB
- Built a **hybrid search system** combining **text, image and metadata** based queries for efficient data exploration
- Integrated perception models like **Grounding DINO**, in-house lane marking and object detection to automate **ground truth generation** using batch workflows with Ray Serve inference
- Enabled **video embedding** generation and search using a **Redundancy-Aware Frame Selection** strategy
- Containerized & deployed on **AWS EKS** with **per-user dataset spaces**, Keycloak auth on Argo endpoints; packaged manifests with Helm, configured Traefik and Route53 for secure, domain-based access
- Added a FiftyOne SDK export/import strategy that **caches datasets** to speed up per-user loading

ML Platform & MLOps Engineering

- Designed a reusable **Ray distributed-training** framework using AWS CDK and on-demand **EKS RayClusters**, integrating ClearML logging and automated node teardown to eliminate idle GPU compute costs
- Established a centralized **CI/CD** automation framework across **24+ core repositories** and deployed an end-to-end GPU/CPU observability stack (OpenTelemetry + Grafana) spanning on-prem, GKE and AWS cloud instances
- Created a GitLab-triggered automated **training-to-deployment pipeline** using AWS Step Functions, SageMaker, Lambda, and EFS/S3; optimized EKS compute nodes using Fargate and EC2 **node-autoscaling**

Perception: Camera 3D Object Detection & LiDAR Auto-Labelling

- Contributed to development of in-house SOTA **monocular 3D object-detection** network (based on [Simonelli et al.](#)) to strengthen the existing camera-only vehicle perception pipeline for Ground Truth Project
- Identified architectural bottlenecks and deployed fixes that increased 2D-to-3D localization accuracy from **55% to 91%**; handled dataset class-imbalance through hyperparameter fine-tuning and data augmentation
- Pytorch tracing based inference pipeline as a standalone API powering the internal **Ground Truth System**, and drove regular dataset annotation quality-check activities across the perception squads
- Built a scalable **auto-labelling pipeline** for LiDAR point-cloud panoptic segmentation, cutting manual annotation effort and accelerating downstream model training
- Benchmarked on **nuScenes** with 6 camera inputs: projected LiDAR points onto 2D panoptic masks generated one-shot by **Segment Anything (SAM)**, reprojected to 3D, and applied kNN to label residual points; scaled batch processing with Ray Serve

AGROSTAR

Pune, India

Machine Learning Intern

May 2019 – Jul 2019

- Developed an Indic-language speaker-diarization pipeline to extract useful labels from call-center recordings, processing **10k+ calls daily**, reducing operations costs by **76.2%** and turn-around time by **~50%**
- Chunked the audio stream via a speech-to-text model and clustered the output with K-Means to identify speakers
- Finetuned an AlexNet-based CNN to filter explicit/NSFW content from the core app feed; engineered data augmentations that optimized the model's threshold, increasing the production **F1-score from 0.67 to 0.70**

LEADERSHIP & MENTORSHIP

- **Knowledge Sharing, JLR**: Led internal sessions for engineering teams on LLMs, VLMs, Kubernetes and MLOps practices, alongside code review and technical mentorship of junior engineers

EDUCATION

INDIAN INSTITUTE OF TECHNOLOGY KANPUR

Kanpur, India

B.Tech – M.Tech (Dual Degree) in Aerospace Engineering | CPI: 7.7/10

Jul 2016 – May 2021

- **Research Publication**: A. Sharma, N. Jain, M. Kothari, pre-print: "[Lightweight Multi-Drone Detection and 3D-Localization via YOLO](#)" (Available on arXiv)
- **Relevant Coursework**: Machine Learning, Linear Algebra, Probability & Statistics, Data Structures & Algorithms

OPEN SOURCE PROJECTS

- **Swarm Drone Dataset Release**: Developed, curated, and publicly released a synthetic drone dataset optimized for multi-agent swarm detection and localization use cases ([Github Repo](#))
- **Student Search App (IIT Kanpur Programming Club)**: Built an Android student-search app with **5k+ downloads** and **4.3★** rating on the Play Store ([App](#), [Code](#))

AWARDS & ACHIEVEMENTS

- **Global Technical Specialist Conference, JLR (2025)**: Selected internationally to present a research poster and PoC on agentic MLOps onsite at JLR UK
- **Creators Kudos & Exceptional Creator Award, JLR (2024)**: Received for outstanding contributions and technical innovation across core enterprise infrastructure projects
- **Global Winner, JLR Hackathon (2023)**: Awarded 1st place worldwide; solution judged best architecture for 360-degree object detection using a camera-only sensor suite
- **JEE Main & Advanced (2016)**: Secured All India Rank **1936** amongst 1.5 million candidates in JEE Main and All India Rank **2925** in JEE Advanced